

Demonstration of Proposed Features

Date	14 July 2026
TeamID	SWUID2026221635
ProjectName	A Comprehensive Measure of Well-Being (HDI Prediction & Analysis)
MaximumMarks	1Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented/ Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Secure Authentication	Login and registration validations handling session access security.	Implemented	Yes	Successfully handles missing fields and invalid login patterns gracefully.
2	Interactive Parameter Sliders	Frontend web inputs using HTML5/CSS3 sliders for custom user inputs.	Implemented	Yes	Provides real-time interactive adjustments for Life Expectancy, Schooling, and GNI.
3	Regression Prediction Model	Python backend utilizing a Scikit-learn Linear Regression model loaded via Pickle.	Implemented	Yes	Computes the continuous numerical index value accurately on the fly.
4	Categorical Tier Classification	Mapping algorithm assigning the calculated continuous score into matching UNDP tiers.	Implemented	Yes	Accurately assigns values into Low, Medium, High, or Very High categories.
5	Visual Chart Representation	Chart generator rendering Matplotlib/Seaborn graphics on the web UI.	Implemented	Yes	Renders static bar charts comparing user inputs side-by-side with target baselines.
6	Dynamic Report Export	Client-side feature creating automated print layouts for	Implemented	Yes	Allows users to successfully

		saving summaries.			export their specific scenario analytics as a local PDF document.
--	--	-------------------	--	--	---

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%